PROCUREMENT PACKAGE:	EL008A - 415V Essential & Emergency MCCs

EQUIPMENT NUMBER(S):		EQUIPMENT NAME(S):	MAIN SUPPLY	BACKUP SUPPLY	ISOLATOR/ TRANSFER SWITCH RATING	BUSBAR RATING	FAULT WITHSTAND CURRENT RATING
1	F12-SW-101	Primary Crushing Essential 415V Switchboard	MCC	Generator	630A	630A	65kA for 1 second
2	F13-SW-102	SAG Mill Essential 415V Switchboard	MCC	F13-SW-103	250A	250A	65kA for 1 second
3	F13-SW-103	Ball Mill Essential 415V Switchboard	MCC	Generator	800A	800A	65kA for 1 second
4	F13-SW-114	Recycle Crushing Essential 415V Switchboard	MCC	Generator	400A	400A	65kA for 1 second
5	F13-SW-120	Mill Chillers & Reagents Essential 415V Switchboard	MCC	Generator	400A	400A	65kA for 1 second
6	F16-SW-104	Flotation Essential 415V Switchboard	MCC	Generator	400A	400A	65kA for 1 second
~~7~~	~~F21-SW-106~~	~~CIL4 Essential 415V Switchboard (SP2)~~	~~MCC~~	~~F21-SW-107~~	~~250A~~	~~630A~~	~~65kA for 1 second~~
~~8~~	~~F21-SW-107~~	~~CIL4 Emergency 415V MCC (SP2)~~	~~MCC~~	~~Generator~~	~~1000A~~	~~960A~~	~~65kA for 1 second~~
9	F22-SW-131	Carbon Treatment & Elution Emergency 415V MCC (SP2)	MCC	Generator	400A	400A	65kA for 1 second
10	F23-SW-123	Final Tails Pumps Emergency 415V MCC	MCC	Generator	630A	630A	65kA for 1 second
11	F28-SW-109	Ultrafine Grinding Essential 415V Switchboard	MCC	Generator	400A	400A	65kA for 1 second
12	F72-SW-172	Reagents Emergency 415V MCC (SP2)	MCC	Generator	630A	630A	65kA for 1 second

REFERENCE DOCUMENTS:		
STAGE	**DOCUMENT NUMBER**	**DESCRIPTION**
Stage 1 /2	2000-F00-LST-EL-11003	415V MCC Module list
Stage 1	2000-F12-SLD-EL-10211	Primary Crushing, F12-SW-101 415V Essential Switchboard, Single Line Diagram
Stage 1	2000-F13-SLD-EL-10311	Milling & Classification – SAG Mill, F13-SW-102 415V Essential Switchboard, Single Line Diagram
Stage 1	2000-F13-SLD-EL-10911	Milling & Classification – Ball Mill, F13-SW-103 415V Essential Switchboard, Single Line Diagram
Stage 1	2000-F13-SLD-EL-10411	Milling & Classification - Recycle Crushing, F13-SW-114 415V Essential Switchboard, Single Line Diagram
Stage 1	2000-F13-SLD-EL-10711	Milling & Classification, F13-SW-120 415V Essential Switchboard, Single Line Diagram
Stage 1	2000-F16-SLD-EL-10251	Rougher and Scavenger Flotation, F16-SW-104 415V Essential Switchboard, Single Line Diagram
~~Stage 2~~	~~2000-F21-SLD-EL-10211~~	~~Flotation Tailings CIL4, F21-SW-106 415V Essential Switchboard, Single Line Diagram~~
~~Stage 2~~	~~2000-F21-SLD-EL-10261~~	~~Flotation Tailings CIL4, F21-SW-107 415V Emergency MCC, Single Line Diagram~~
Stage 2	2000-F22-SLD-EL-10251	Carbon Regen & Elution, F22-SW-031 415V Emergency MCC, Single Line Diagram
Stage 1	2000-F23-SLD-EL-10211	Final Tails Handling and Storage, F23-SW-123 415V Emergency MCC, Single Line Diagram, Sheet 1
Stage 1	2000-F23-SLD-EL-10212	Final Tails Handling and Storage, F23-SW-123 415V Emergency MCC, Single Line Diagram, Sheet 2
Stage 1	2000-F28-SLD-EL-10211	Ultra Fine Grinding 2/3, F28-SW-109 415V Essential Switchboard, Single Line Diagram
Stage 2	2000-F72-SLD-EL-10251	Reagents, F72-SW-172 415V Emergency MCC, Single Line Diagram
Standard drawing	2000-F00-STD-EL-10102	415V Fxx-SW-xxx 24VDC Distribution Schematic and Termination Diagram
Standard drawing	2000-F00-STD-EL-10103	415V Fxx-SW-xxx 0.37..30kW DOL Pump Schematic and Termination Diagram

REFERENCE DOCUMENTS:		
STAGE	**DOCUMENT NUMBER**	**DESCRIPTION**
Standard drawing	2000-F00-STD-EL-10104	415V Fxx-SW-xxx 37..75kW DOL Pump Schematic and Termination Diagram
Standard drawing	2000-F00-STD-EL-10105	415V Fxx-SW-xxx 90..220kW DOL Pump Schematic and Termination Diagram
Standard drawing	2000-F00-STD-EL-10106	415V Fxx-SW-xxx 250kW+ DOL Pump Schematic and Termination Diagram
Standard drawing	2000-F00-STD-EL-10107	415V Fxx-SW-xxx 0.37..30kW VSD Pump Schematic and Termination Diagram
Standard drawing	2000-F00-STD-EL-10108	415V Fxx-SW-xxx 37..75kW VSD Pump Schematic and Termination Diagram
Standard drawing	2000-F00-STD-EL-10109	415V Fxx-SW-xxx 90..200kW VSD Pump Schematic and Termination Diagram
Standard drawing	2000-F00-STD-EL-10110	415V Fxx-SW-xxx 250kW+ VSD Pump Schematic and Termination Diagram
Standard drawing	2000-F00-STD-EL-10111	415V Fxx-SW-xxx 0.37..30kW DOL Sump Pump Schematic and Termination Diagram
Standard drawing	2000-F00-STD-EL-10112	415V Fxx-SW-xxx 0..37..30kW DOL Conveyor Schematic and Termination Diagram
Standard drawing	2000-F00-STD-EL-10113	415V Fxx-SW-xxx 37..75kW DOL Conveyor Schematic and Termination Diagram
Standard drawing	2000-F00-STD-EL-10114	415V Fxx-SW-xxx 90..220kW DOL Conveyor Schematic and Termination Diagram
Standard drawing	2000-F00-STD-EL-10115	415V Fxx-SW-xxx 0.37..30kW VSD Conveyor Schematic and Termination Diagram
Standard drawing	2000-F00-STD-EL-10116	415V Fxx-SW-xxx 37..75kW VSD Conveyor Schematic and Termination Diagram
Standard drawing	2000-F00-STD-EL-10117	415V Fxx-SW-xxx 90..200kW VSD Conveyor Schematic and Termination Diagram
Standard drawing	2000-F00-STD-EL-10118	415V Fxx-SW-xxx 250kW+ VSD Conveyor Schematic and Termination Diagram
Standard drawing	2000-F00-STD-EL-10119	415V Fxx-SW-xxx 0.37..30kW Reversing DOL Drive Schematic and Termination Diagram
Standard drawing	2000-F00-STD-EL-10120	415V Fxx-SW-xxx 0.37..30kW DOL Dual Motor Screen Schematic and Termination Diagram
Standard drawing	2000-F00-STD-EL-10121	415V Fxx-SW-xxx 0.37..30kW VSD Dual Motor Feeder Schematic and Termination Diagram
Standard drawing	2000-F00-STD-EL-10122	415V Fxx-SW-xxx 30..300A Active Harmonic Filter Schematic and Termination Diagram
Standard drawing	2000-F00-STD-EL-10124	415V Fxx-SW-xxx 20..250A Switchroom DB Feeder Schematic and Termination Diagram
Standard drawing	2000-F00-STD-EL-10125	415V Fxx-SW-xxx 20..800A MCCB Feeder Schematic and Termination Diagram
Standard drawing	2000-F00-STD-EL-10126	415V Fxx-SW-xxx 20..400A Decontactor Feeder Schematic and Termination Diagram
Standard drawing	2000-F00-STD-EL-10127	415V Fxx-SW-xxx DOL Heater Schematic and Termination Diagram
Standard drawing	2000-F00-STD-EL-10128	415V Fxx-SW-xxx 220..250kW VSD Pump Schematic and Termination Diagram
Standard drawing	2000-F00-STD-EL-10129	415V Fxx-SW-xxx 90..220kW Reversing DOL Drive Schematic and Termination Diagram

REFERENCE SPECIFICATIONS:

1. 2000-F00-STS-EL-10014_2 - Preferred Equipment Suppliers for Electrical, Instrumentation And Controls Specification
2. 2000-F00-STS-EL-10016_1 - Low Voltage Switchboards & Switchgear Electrical Specification
3. 2000-F00-STS-GE-10002_1 - Site Specific Information
4. 2000-F00-STS-GE-10003_3 - Plant Numbering Specification
5. 2000-F00-STS-ST-10001_3 - Steelwork Fabrication
6. 2000-F00-STS-ST-10002_2 - Steelwork Erection Specification
7. 2000-MEP-EL008-SCH1 - Vendor Document Schedule
8. SPEC-00-G-05_12 – Technical Specification for Protective Coatings

SCOPE OF WORK

Work Included

Essential Switchboards:

- Design, supply and fabrication of seven (7) indoor single-sided 415V switchboards including, including but not limited to:
  - Incomer with two (2) off full load isolator Switches with and one (1) off Manual Transfer Switch (MTS);
  - Demountable Form 4a Outgoing Modules;
  - All internal Switchboard Components including circuit breakers, E/L relays, contactors, overload relays, 24V DC Distribution, terminals, shrouds etc;
  - Switchboard shall be fitted with VAMP Arc Fault I/O modules for arc fault detection. Optical arc fault detection shall be provided at incomer and all outgoing modules including cable ways. An arc fault detected on the Switchboard shall trip the upstream Mains Supply MCC incomer. The VAMP Arc Fault I/O modules shall be wired back to the Main Supply MCC's Arc Fault I/O modules chain and reports to the Main Supply MCC's Arc Fault Controller.
  - All internal wiring and cabling including power, control, copper communications to connect to switchroom DCS panel;
  - Labelling, including stainless steel equipment nameplate;
  - Lifting lugs, as applicable;
  - Any other items which are normally supplied with this equipment;
  - 20% spare tier space including but not limited to terminals, gland plate size, control power supply sizing, switchboard capacity;
- Surface Protection;
- All ancillary items and special tools normally supplied with, or required to adjust and maintain all components associated with the equipment;
- All fasteners, gaskets, packers, shims and any other items necessary for site assembly;
- Factory Acceptance Testing at manufacturer's premises;
- Supply of all necessary documentation, drawings and test documentation;

Emergency MCCs:

- Design, supply and fabrication of three (3) indoor single-sided 415V motor control centres (MCCs), including but not limited to:
  - Incomer with two (2) off full load isolator Switches with and one (1) off Auto Transfer Switch (ATS);
  - Demountable Form 4a Outgoing Modules;
  - All internal MCC Components including circuit breakers, E/L relays, contactors, overloads relays, 24VDC Distribution, terminals, shrouds etc;
  - Installation of free issue items into top of relevant tiers
    - Cisco network switches installed in the top of tier cable zones to marshal intelligent electronic devices (IEDs) such as EOLs and relays. Maximum of 1 per two tiers, vendor to confirm required quantity of ports.
  - CAT6 wiring to IEDs (EOLs, relays, etc).
  - MCC shall be fitted with VAMP Arc Fault I/O modules for arc fault detection. Optical arc fault detection shall be provided at incomer and all outgoing modules including cable ways. An arc fault detected on the MCC shall trip the upstream Mains Supply MCC incomer and Genset (onboard) circuit breaker. The VAMP Arc Fault I/O modules shall be wired back to the Main Supply MCC's Arc Fault I/O modules chain and reports to the Main Supply MCC's Arc Fault Controller.
  - All internal wiring and cabling including power, control, copper communications and fibre communications;
  - Labelling, including stainless steel equipment nameplate;
  - Lifting lugs, as applicable;
  - Any other items which are normally supplied with this equipment;
  - 20% spare tier space including but not limited to terminals, gland plate size, control power supply sizing, switchboard capacity;
  - Extendable one end.
- Surface Protection – manufacturer's standard;
- All ancillary items and special tools normally supplied with, or required to adjust and maintain all components associated with the equipment;
- All fasteners, gaskets, packers, shims and any other items necessary for site assembly;
- Factory Acceptance Testing at manufacturer's premises;
- Supply of all necessary documentation, drawings and test documentation;

Special Notes:

Nil

Work Excluded

- Cabling external to switchboard and switchroom;
- Spare parts;
- Installation and commissioning at site.

Design Notes

EPC·S document package includes MCC Modules List and typical schematics. Design has been described using typical schematics. Actual schematics will be produced shortly following award of contract, if successful. 24V DC distribution schematic will be updated to suit MCC GA following receipt of successful vendor's approved MCC GA drawings.

Switchboard or MCC incomer protections to be provided by the upstream Main Supply MCC's feeder Circuit Breaker, Backup Power MCC's feeder Circuit Breaker or the Generator's onboard circuit breaker. The Switchboard or MCC incomer shall be provided with full load isolator switches and a 3 position transfer switch (Manual or Auto as specified).

Supplier shall nominate suitable circuit breaker sizes to suit the type 2 high efficiency coordination tables for the offered switchgear. Multi-motor drives shall ensure that electronic overloads and their associated contactors are type 2 high efficiency coordinated to suit the individual motors and shall be suitable for the full load current, noting that vibrating motors (such as those used in screens) typically have a larger full load current than a standard motor.

DOL motor starter modules will utilise electronic overloads with Ethernet communications functionality. Protection functions required from the overload are stated further in this datasheet. Soft Starters, where utilised, will have ethernet communications. IO Expansion modules may be utilised where required (E.g. multi-motor DOL).

VSD feeder modules to be fitted with current limiting fuses where it is required to achieve adequate protection to the cable and/or the VSD.

24V DC distribution schematic describes indicative 24V DC distribution throughout the MCC. 24V DC power will be fed from the Main Supply MCC's 24VDC Power Supplies Units. It is envisaged that an MCB will be installed at the top of each cable zone. 24V DC control supply will be looped between DC distribution compartment and MCBs at top of tier. Within each tier, 24V DC control will loop from the MCB at top of tier to each individual starter module. Supplier shall ensure that their offer includes all necessary equipment for the 24V DC distribution.

Live line indication and earth leakage shall be provided for all incomers and Functional Unit Modules. Other forms of imbalance or earth leakage protection required for all hardwired sub-circuits.

Arc fault detection shall be provided at incomer and all outgoing modules including cable ways via optical sensors. The arc fault detection shall trip the upstream Main Supply MCC incomer when on mains power, and the Generator (onboard) circuit breaker when on auto transferred backed up power.

The key differences in Essential Switchboard and Emergency MCC designs are the Essential Switchboards utilise a manual transfer switch and do not require installation of free-issued network switches. The Emergency MCCs utilise an ATS and will require installation of free-issued CISCO network switches in cable zones.

DUTY SPECIFICATION	**UOM**	EPC **DATA** VTA = Vendor to Advise	**SUPPLIER DATA** Supplier to Complete blank cells and indicate acceptance or otherwise as appropriate	**NOTES**
General				
Equipment Number	-	Refer above		1
Equipment Name	-	Refer above		1
Make	-		POWINS	2
Model	-		ECO LOCK PREMIUM	2
MCC Type	-	Air Insulated, Metal Clad, Form 4a, Free Standing, Double Sided		1
Non-ACB Functional Unit Connections (AS61439)	-	Demountable	Confirmed	1
Country of Manufacture	-		Malaysia/Perth Australia	2
Compliance with Australian Standards	Yes/No	Yes – AS3439, AS3947, AS60947, AS61439	Confirmed	1, 6
Drive and Load Descriptions	-	As per MCC Module List – 2000-F00-LST-EL-11003	Confirmed	1
Environmental Conditions				
Plant Location	-	Western Australia		1
Service Location	Indoor/Outdoor	Indoor		1
Ambient Indoor Air Temperature - Min	°C	21		1
Ambient Indoor Air Temperature - Max	°C	27		1
Humidity	%	0 to 99		1
Altitude	m	<1000m		1
Supply				
Supply	-	3P + N + E		1
Input Voltage	V	415V +/- 10%		1
Frequency	Hz	50 (±2%)		1

DUTY SPECIFICATION	UOM	EPC DATA VTA = Vendor to Advise	SUPPLIER DATA Supplier to Complete blank cells and indicate acceptance or otherwise as appropriate	NOTES
Neutral Earthing	-	Solid; M.E.N. on Main Supply MCC Ref. AS3010 Figure 2.5		1
Switchboard Ratings				
Type Tested Assembly to AS61439	No/Partially/Fully	Fully Type Tested Assembly	Confirmed	1, 6
Switchboard Rating Plate	-	Yes	Confirmed	1
Main Bus Bar Continuous Current Rating	A	Refer to Table above. Bus shall be fully rated along its length. Supplier to confirm ratings for site conditions.	Confirmed	3
Neutral Bus Bar Continuous Current Rating	A	100% of phase rating	Confirmed	3
Main & Neutral Bus Bars Fault Withstand Level	kA for sec	Minimum 65kA for 1 Second	Confirmed	3
Dropper Bus Bars Fault Withstand Level	kA for sec	Minimum 65kA for 1 Second	Confirmed	3
Earth Bus Bars Fault Withstand Level	kA for sec	VTA	Confirmed	2
Rated Insulation Voltage	V	1000V	Confirmed	3
Interruption Capacity (All circuits connected directly and indirectly to main bus Including ACBs, MCCB and 240V control circuit protection). ** Cascading is not permitted, fault current limiting fuses to be used where required.	kA	65kA Icu Vendor to Advise kA Ics	Confirmed	3
Incomer Current Rating (de-rated according to manufacturer's de-rating tables and enclosure)	A	Refer to table above	Confirmed	3
Internal Arc Fault Containment (IAC) Rating to IEC TR 61641 / AS 61439.1 Annex ZD - Incomer Line Side Terminals - Main & Dropper Bus Bar System - Outgoing Modules	kA for sec	Supplier to confirm ratings below: 1. 63kA/0.3s minimum 2. IAC rated bus not required if insulated 3. 63kA/0.3s minimum	Confirmed. Outgoing modules to AS61439.1 Annex ZD	3
Cascading used to achieve minimum breaking capacity	Yes / No	No		1
Switchboard Construction				
General Construction	-	Free Standing Metal Clad Air Insulated	Confirmed	3
Plinth Construction	-	VTA	HDG Steel	2
Degree of Ingress Protection	IP	VTA – Min IP42	Confirmed	3
Type of Cooling	-	Air cooled (arc contained non ventilated)	Confirmed	1
Exterior Paint Colour	-	AS2700 Tropical Blue T11	Confirmed	1
Interior Paint Colour	-	AS2700 / White	Confirmed	1
Paint Specification	-	Vendor Standard	Confirmed	2
Single Sided / Double Sided	Single / Double	Single	Confirmed	1
Spare Space Allowance for Future Outgoing Modules		Essential: 20% spare space Emergency: 20% spare space	All switchboards & MCC's have 20% spare space minimum except for F21-SW-107 and F28-SW-109 which both have 16% spare space. Some switchrooms do not have available space for a future tier	3
Capacity for Future Extension	-	Essential: No Emergency: Yes	Confirmed	1
Switchboard Incomer Form	-	Form 4a	Confirmed	1
Switchboard Outgoing Motor Starters and Feeders Form	-	Form 4a	Confirmed	1
Cable Entry	Top / Bottom	Bottom Entry	Confirmed	1
Drop Boxes	Yes/No	Yes, with side panels to allow single person glanding of cables from below (part of switchroom contract)	Confirmed	1
Gland Plates	-	5mm Aluminium, earthed with neoprene gasket	Confirmed	1

DUTY SPECIFICATION	UOM	EPC DATA VTA = Vendor to Advise	SUPPLIER DATA Supplier to Complete blank cells and indicate acceptance or otherwise as appropriate	NOTES
Lifting Lugs	-	VTA	Lifting from plinth only	2
Jacking Beam Lugs	-	No – Base of switchboard must be vermin proof and IP rated.	"Switchroom gland plates form	1
Busbars				
Phase Busbar Size	mm^2	VTA	630A Bar: 1 x 60 x 10 960A Bar: 1 x 100 x 10	2
Neutral Busbar Size	mm^2	VTA	630A Bar: 1 x 60 x 10 960A Bar: 1 x 100 x 10	2
Earth Busbar Size	mm^2	VTA	1 x 60 x 6	2
Phase Dropper Size	mm^2	VTA	1 x 60 x 6 as minimum	2
Neutral Dropper Size	mm^2	VTA	1 x 60 x 6 as minimum	2
Busbar Identification	-	Colour Coded (Red, White, Blue Phases, Black Neutral & Green/Yellow Earth)	Confirmed	1
Busbar Material	-	HDHC Copper	Confirmed	1
Fully Insulated Bus Bars and Droppers	Yes / No	Yes	Confirmed	1
Main Busbar Location / Accessibility	-	Top - Front accessible IP6X metal clad segregation	Confirmed	3
Phase and Wire Colours				
Phase Colours	-	Red, White, Blue	Confirmed	1
Neutral Colour	-	Black	Confirmed	1
Earth Colour	-	Green / Yellow	Confirmed	1
24V DC +24V DC (+ve)	-	Orange	Confirmed	1
24V DC 0V (-ve)	-	Purple	Confirmed	1
48V DC Tripping and closing (+ve) – HV Swgr	-	Grey	Confirmed	1
48V DC Tripping and closing (-ve) – HV Swgr	-	Black with white tracing	Confirmed	1
Analog 4-20mA	-	Pink	Confirmed	1
Ethernet cables	-	Grey Sheath	Confirmed	1
Fibre optic patch leads	-	Yellow – Single Mode	Confirmed	1
Live Line Indication				
Manufacturer	-	Nilsen	Confirmed	3
Model	-	NWA2019	Confirmed	3
Incomer Cubicle	-	Yes – Indicator mounted on control compartment	Confirmed	1
Incomer Cubicle Live Line Sensing	-	Pick up from line side of ACB/MCCB, as per schematic	Confirmed	1
Outgoing Modules	-	Yes – Indicator mounted in module door. Pick up from downstream side of MCCB, as per schematics	Confirmed	1
Locking				
Incomer isolator switch padlockable in OFF position	Yes / No	Yes - 6.3mm shackle	"Confirmed On Manual Transfer MCCB door hands and on the motor operator of the ATS MCCB's "	1
Transfer switch padlockable in OFF position	Yes / No	Yes - 6.3mm shackle	"Interlocked MCCB's offered. MCCB changeover only. No additional isolator "	1
Outgoing MCCB come with mechanically interlocked cubicle door mounted rotary handles so that: * The door cannot be opened in the on position. * The circuit breaker rotary handle can be locked in the off position and the door cannot be opened while padlocked off. * The circuit breaker extension handle shall be lockable inside the module allowing the MCCB to be locked out while the door is open. * The circuit breaker rotary handle cannot be padlocked in the on position.	Yes / No	Yes - 6.3mm shackle	Confirmed	1

DUTY SPECIFICATION	UOM	EPC DATA VTA = Vendor to Advise	SUPPLIER DATA Supplier to Complete blank cells and indicate acceptance or otherwise as appropriate	NOTES
Incomer Switch				
Incomer Type	-	Load break switches rated category AC-23B severe load duty.	Interlocked MCCB's only offered in place of Isolator and the changeover switch	1
Fixed / Demountable / Withdrawable	-	Fixed	Confirmed	1
Incomer Switch Make		VTA	Terasaki Tembreak Pro	2
Incomer Switch Model	-	VTA	Terasaki Tembreak Pro	2
Incomer Switch Rating	-	Refer to Table above.	Confirmed	3
Surge Diverters				
Surge Diverters required on Incomer	Yes/No	Yes; Backup Power Incomer Switch only	Confirmed	1
Manufacturer	-	VTA	Novaris	2
Model	-	VTA	SD3-200-275	2
Minimum and Maximum Fuse Size	-	VTA	63A	2
Provides Surge Diverter Status Contact to DCS	Yes / No	Yes	Confirmed	1
Auto Transfer Switch – Emergency MCC				
Manufacturer	-	Socomec	Tembreak Pro MCCB. See above	1
Model	-	ATyS P		1
Transfer Switch Rating	-	Refer to Table above.		3
Current Sensing	Yes / No	No		1
Voltage Sensing	Yes / No	yes		1
Functionality	-	Monitor via Voltage Sensing. <u>Loss of Main:</u> Auto Transfer to Generator Supply, remote start generator. <u>Main resume:</u> Auto Transfer to Main Supply after a 5 minutes delay, remote stop generator.	Auto Changeover controller offered is the Socomec Atys C65 to control the MCCB's	1
Remote Display	Yes / No	Yes; ATyS D20	Not offered. See data sheet for C65 controller	1
Communication	Yes / No	Yes; Ethernet Communication module (P/N: 48250203)	Not offered. See data sheet for C65 controller	1
Manual Transfer Switch – Essential Switchboard				
Manufacturer	-	Socomec	Tembreak Pro MCCB. See above	1
Model	-	SIRCOVER		1
Transfer Switch Rating	-	Refer to Table above.		3
Functionality	-	Manual 3-position, make before breaker change-over		1
Phase Monitoring Relay – Essential Switchboard				
Required	Yes / No	Yes, with exception of: F13-SW-102 F21-SW-106 F21-SW-107 F22-SW-131 F23-SW-123	Confirmed	1
Manufacturer	-	Schneider	Confirmed	1
Model	-	RM17TG00	Confirmed	1
Rating	-	415VAV 50Hz	Confirmed	1
Functionality	-	Monitor phase sequence and phase loss; Wired to a local alarm indicator to provide visual warning before manual switching of transfer switch.	Confirmed	1
Current Transformers				
Current Transformers	Yes / No	No	Confirmed	1
Arc Fault Detection System				
Manufacturer	-	Schneider	Confirmed	1
I/O Module Model	-	VTA	Confirmed	3
Power Supply	-	24Vdc	Confirmed	1

DUTY SPECIFICATION	UOM	EPC DATA VTA = Vendor to Advise	SUPPLIER DATA Supplier to Complete blank cells and indicate acceptance or otherwise as appropriate	NOTES
Communications	-	Via Modbus TCP Communication module	Confirmed	1
Design note	-	Supplier to design, perform product selection, install and commission Arc Fault Detection optical sensors and I/O modules to interface with the VAMP321 located in Switchroom main MCC. No VAMP321 required on Emergency MCC or Essential Switchboard.	Confirmed	3, 7
Optical Sensor Locations	-	Incomer and all outgoing modules including cable ways. Arc fault monitoring are not required for areas (E.g. busbar) where the component design is classified as an arc free zone in accordance to IEC61641.	Confirmed. Bus bar chamber and cable zones do not have arc sensors	1
VSD Feeders				
Feeder Type	-	Moulded Case Circuit Breaker (MCCB)	Confirmed	1
Fixed / Demountable / Withdrawable	-	Demountable	Confirmed	1
MCCB Make	-	VTA	Terasaki	2
MCCB Model (series)	-	VTA	Tembreak Pro	2
MCCB Protection	-	Adjustable Magnetic as standard, rating as per manufacturer's protection table	Confirmed	1
Outgoing cable connection (cable(s) exiting MCC)	-	VTA - Onto MCCB or terminals as per manufacturer's recommendation	Confirmed	1
Neutral Connection	-	Provided inside feeder module via Neutral busbar dropper	Confirmed	1
VSD Starter Control Voltage	-	24V DC	Confirmed	1
DOL Motor Starters				
Feeder Type	-	Moulded Case Circuit Breaker (MCCB)	Confirmed	1
Fixed / Demountable / Withdrawable	-	Demountable	Confirmed	1
MCCB Make	-	VTA	Terasaki	2
MCCB Model (series)	-	VTA	Tembreak Pro	2
MCCB Protection	-	Adjustable Magnetic as standard, rating as per manufacturer's Type 2 coordination chart for IE3 high efficiency motors. 3 Pole with Solid Neutral sensing, adjustable earth leakage and electronic trip units as shown on Schematics.	Standard 3 pole Thermal Magnetic circuit breaker offered. Earth Leakage provided by NHP relay	1
Contactor Model	-	Type 2 Coordinated AC3 duty suitable for IE3 high efficiency motors	Confirmed	3
Electronic Overload Relay (EOL)				
Electronic Overload Make / Model	-	Rockwell E300 series	Confirmed	1
Electronic Overload functions		As a minimum: - Current sensing - Class 10 motor thermal protection - I/O modules (If required) - PTC Thermistor input	Confirmed. Model included does not have Earth Leakage or Ground Fault protection as this is provided by the Earth Leakage relay	3

DUTY SPECIFICATION	UOM	EPC DATA VTA = Vendor to Advise	SUPPLIER DATA Supplier to Complete blank cells and indicate acceptance or otherwise as appropriate	NOTES
Electronic Overload Communication		Modbus TCP – Anybus-E300-MBTCP Comms Module	Confirmed	1
Electronic Overload Expansion I/O module		Where required	As per Schematics	1
Sensing module		Integral or External to suit motor duty	Confirmed	1
HMI/ Starter Control station module		Required	Confirmed	1, 7
Thermal Overload Model (applicable to multi-motor DOL Starters)	-	VTA - Selected to suit Type 2 coordination tables for AC3 duty for high efficiency motors	Confirmed	3
Outgoing cable connection	-	Onto Contactor, terminals, or TOL as per manufacturer's design drawings	Confirmed	1
Neutral Connection	-	Provided inside feeder module via Neutral busbar dropper	Confirmed	1
Motor Starter Contactor Coil Voltage	-	240V AC	Confirmed	1
Motor Starter Control Voltage	-	24V DC	Confirmed	1
Soft Starter Motor Starters				
Feeder Type	-	Moulded Case Circuit Breaker (MCCB)	N/A	1
Fixed / Demountable / Withdrawable	-	Demountable	N/A	1
CFS Make	-	VTA	N/A	2
CFS Model	-	VTA	N/A	2
Fuse Type/curve	-	HRC, gR curve, 500V AC, Combined Indicator, Insulated grip lugs.	N/A	1
Contactor Model	-	VTA - Type 2 Coordinated AC3 duty suitable for high efficiency motors	N/A	2
Soft Starter Make / Model	-	VTA - with Modbus TCP Communication module, HMI mounted on MCC cabinet door.	N/A	3
Soft Starter Location	-	Inside MCC cubicle.	N/A	1
Outgoing cable connection	-	Onto soft starter terminals as per design drawings	N/A	1
Neutral Connection	-	Provided inside feeder module via Neutral busbar dropper	N/A	1
Contactor Coil Voltage	-	240V AC	N/A	1
Soft Starter Control Voltage	-	24V DC	N/A	1
MCCB Feeders				
Feeder Type	-	Moulded Case Circuit Breaker (MCCB) with Earth Leakage	Confirmed	1
Fixed / Demountable / Withdrawable	-	Demountable	Confirmed	1
Feeder Make		VTA	Terasaki	2
Feeder MCCB Model	-	VTA	Tembreak Pro	2
MCCB Protection	-	Adjustable Thermal / Magnetic as standard, 3 Pole with Solid Neutral sensing, adjustable earth leakage and electronic trip units where shown on SLD. VTA range of adjustment offered for thermal/long time trip.	Standard 3 pole Thermal Magnetic circuit breaker offered. Earth Leakage provided by NHP relay	3
Fault limiting fuses	-	Yes, where specified on standard drawing.	Confirmed	1
Outgoing cable connection	-	Onto terminals as per manufacturer's design drawings	Confirmed	1
Neutral Connection		Provided inside feeder module via Neutral busbar dropper	Confirmed	1
Standard Motor Starter and VSD Feeder Sizes				

DUTY SPECIFICATION	UOM	EPC DATA VTA = Vendor to Advise		SUPPLIER DATA Supplier to Complete blank cells and indicate acceptance or otherwise as appropriate	NOTES
DOL Motor starters, VSD feeders and soft starter motor starters grouped into standard sizes for each starter type with all starters of the same type and size using the same components	Yes / No	Yes		Confirmed	1, 5
Nominated standard sizes	-	Vendor to advise standard module sizes, EPC standard schematics will be adjusted to suit		TBA	3, 5
Control Power Supply & Distribution					
24V DC distribution	-	24VDC Power fed from Switchroom Main Supply MCC's 24VDC distribution.		Confirmed. Power Supply is external to the MCC	1
Separate 24V DC circuit breakers for each tier	Yes / No	Yes – Located at top of cable zone		Confirmed	1
Inter-Panel Wiring					
Routing	-	Routed in cable duct running along top of MCC tiers.		Confirmed	3
Segregation	-	Ethernet Comms to be segregated by metallic barrier or in separate duct.		Confirmed	3
Duct Space for Future Expansion	-	Minimum 20% space to be provided in control wiring ducts for connection of future Functional Unit Modules.		Confirmed	1
Communication					
Network Switch (free issued)		<u>Essential</u>: No	<u>Emergency</u>: Yes, installed in the top of cable zone tiers. Maximum of 1 per every two tiers.	Confirmed	2
Labelling					
Switchboard Label	Etched traffolyte label, located on incomer tier, with Switchboard number, Switchboard description and SLD number at minimum.			Confirmed	1
Switchboard Nameplate Label	Etched traffolyte nameplate label located on incomer tier. Label shall include the switchboard ratings, serial number, and date of manufacture.			Confirmed	1
Tier Label	Etched traffolyte label (black on white background) to be located on the top of each tier, with the Switchboard tier number.			Confirmed	1
Module Outer Door Label	Etched traffolyte label (black on white background) on the outer door of each Switchboard module. Each label shall contain the module tier position, drive kW rating or feeder Amp rating, equipment tag number, equipment description and the Switchboard schematic drawing number.			Confirmed	1
Module Inner Labelling	Etched traffolyte label (black on white background) with equipment number for equipment internal to module (e.g. MCCB, MCB, Relays, Terminal Strips).			Confirmed	1
Terminals and Panel Wiring	Terminals to be numbered per design drawings, internal wiring shall be marked at both ends as per design drawings.			Confirmed	1
Misc Components					
Miniature Circuit Breakers Make & Model	-	VTA		NHP DIN. 10kA Type	2
MCCB Handles	Yes / No	Yes – Interlocked with cubicle doors. Padlockable both inside and outside of module door.		Confirmed	1

DUTY SPECIFICATION	UOM	EPC DATA VTA = Vendor to Advise	SUPPLIER DATA Supplier to Complete blank cells and indicate acceptance or otherwise as appropriate	NOTES
Control Relays	-	VTA Make/Model DC Relays come with integral surge suppression diodes	NHP Finder Type	2
Control and Power Terminal Make & Model	-	Weidmuller WDU series	Confirmed	3
Earth Terminal Make and Model	-	Weidmuller WPE series	Confirmed	3
LED Indicator Lamps & Pushbuttons	-	IPD P9 series	IPD or NHP	3
Voltage test terminals (MCC incomer cubicle)	-	Weidmuller WTL series	Confirmed	3
Earth Leakage Relay – VSD Feeder	Make / Model	AMPCONTROL / ELV	Confirmed	1, 7
Earth Leakage Relay – All other motor starter/ Feeder	Make / Model	NHP / RD1EP21H	Confirmed	1, 7
Test Requirements				
Type (Verification)	**Required,** Provide test certificates		Confirmed	1
Routine	**Required,** This includes: - Insulation Test of MCC bus - Functional checks of all Functional Unit Modules - Incomer (Transfer Switch) function testing - Primary injection testing of 132+ kW DOL external CTs		Confirmed	1
Shipping and Delivery				
Packaging	Installed in switchroom			1
Dimensions and Weight	Vendor to complete **Table A**			2

Notes:

1. Data supplied by EPC The Supplier shall comply with this data.
2. Data shall be provided by Supplier.
3. Data proposed by EPC The Supplier shall confirm data or state alternative.
4. (Deleted)
5. Vendor to nominate standard module sizes.
6. Vendor to provide at tender stage an initial AS61439.1 Appendix D.1 Statement of Compliance (SoC) document for the complete assembly with final SoC provided after testing.
7. Where provided, EOL HMI, Arc flash protection relay, power meter, Earth Fault Relays or Earth Leakage Relays shall be door mounted to access push buttons, dip-switch settings and to view indications.

Name:	

Table A ESSENTIAL SWITCHBOARD AND EMERGENCY MCC Dimensions And Weight

Vendor to complete below table.

NO.	EQUIPMENT NUMBER	LENGTH (mm)	WIDTH (mm)	HEIGHT (mm)	SHIPPING SECTION DIMENSIONS L x H x D (mm)	HEAVIEST SHIPPING SECTION WEIGHT (kg)
1	F12-SW-101	2460	700	2329		
2	F13-SW-102	1660	700	2329		
3	F13-SW-103	2660	700	2329		
4	F13-SW-114	1160	700	2329		
5	F13-SW-120	2460	700	2329		
6	F16-SW-104	2460	700	2329		
~~7~~	~~F21-SW-106~~	~~4320~~	~~700~~	~~2329~~		
~~8~~	~~F21-SW-107~~	~~2660~~	~~700~~	~~2329~~		
9	F22-SW-131	2460	700	2329		
10	F23-SW-123	4320	700	2329		
11	F28-SW-109	1660	700	2329		
12	F72-SW-172					